Hochschule
München
University of
Applied Sciences

Fakultät für Informatik und Mathematik

Pattern Recognition & Machine Learning

Prof. Dr. Markus Friedrich

Challenges

Challenge 1: 3D Scan-to-CAD Alignment for Robotic Part Localization

Challenge Giver	ROBOTe Industrial GmBH (Prof. Dr. Sebastian Pflaum)

In the AutoFIN research project, a system is being developed to automate robotic grinding processes for large pressing tools in automotive body construction. For the robot to process parts autonomously, it must first determine the position and orientation of the part within the robot cell. A solution involves capturing the part with a stationary 3D scanner and aligning the generated point cloud with the existing CAD model of the tool. This alignment allows for the calculation of the part's pose relative to the robot coordinate system.

Initial Requirements

The goal of the project group is to develop a software module that algorithmically implements this automated part localization process based on 3D scan data. This requires investigating and implementing methods for point cloud processing, registering scan data with CAD models, and calculating the transformation matrix between the part and robot coordinate systems. The developed software will form a central component for the automated measurement of parts in robot cells.

- Implementation of algorithms to filter noise, downsample point clouds, and segment relevant areas of the part to prepare the data for registration.
- Investigation of suitable methods for 3D matching between scan data and CAD models. Research and comparison of algorithms and libraries (e.g., OpenCV, Open3D, PCL), specifically focusing on ICP-based methods and other robust registration techniques.
- Calculation of the necessary transformation matrices between the part coordinate system, the scanner coordinate system, and the robot coordinate system, and validating the calculated pose.
- Comprehensive Evaluation: Precision, stability and computational efficiency
- **Expected Outcome:** A functional prototype capable of reading and processing 3D scan data, registering scan data with a CAD model, calculating the part's pose relative to the robot, and providing results for integration into further software modules (e.g., for robotic path planning).

Starting Points

OpenCV, Open3D, PCL

Challenge 2: Image-Based Assessment of Grinding States

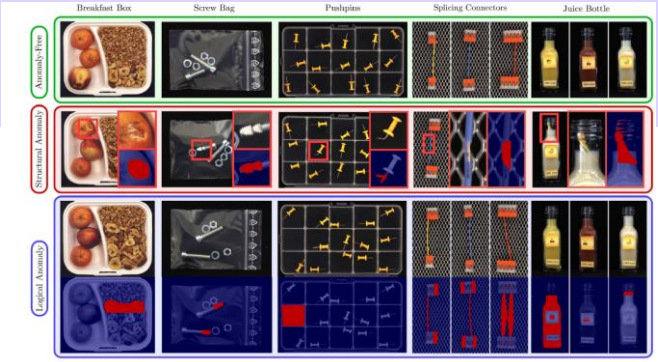
Challenge Giver	ROBOTe Industrial GmBH (Prof. Dr. Sebastian Pflaum)
In the AutoFIN research project, an AI-based system is being developed to evaluate surface conditions during robotic grinding. Precisely classifying the state of metallic surfaces is crucial for automated quality control within the robot cell. An initial YOLOv11 model serves as a benchmark, but now the goal is to investigate whether alternative state-of-the-art architectures offer higher accuracy or better efficiency.	
Initial Requirements	The project' goal is to train and systematically compare various machine learning models for classifying grinding states. The aim is to identify the optimal balance between classification accuracy, robustness, and computational effort in order to derive a recommendation for real-world use in the automated process. • Investigation of the class structure and data distribution of the existing dataset (~5,000 images). Application of expansion strategies such as data augmentation to increase model robustness. • Research and implementation of various architectures, such as ResNet, DenseNet, alternative YOLO variants, or Vision Transformers • Execution of systematic training runs using suitable strategies (e.g., transfer learning, learning rate scheduling). • Evaluation of the models against the existing benchmarks (classification performance, efficiency, robustness (under varying image conditions). **Expected Outcome:** A well-founded recommendation for the highest-performing model, which serves as the basis for integration into the automated robot cell.
Possible Starting Points	• **Dataset:** Existing image pool of ground metallic surfaces (AutoFIN project). • **Architectures:** YOLOv11 (Benchmark), ResNet, DenseNet, Vision Transformer (ViT).

Challenge 3: Semantic Inspector – Logical Anomaly Detection

Challenge Giver	MVTec (Ulf Schulmeyer)
This challenge focuses on MVTec LOCO AD, where anomalies are not physical damages (scratches/dents) but violations of structural rules. Teams will implement a Vision-Language Model (VLM) pipeline that autonomously learns "normality" from a few-shot set of images and identifies logical contradictions through automated Question Answering (QA).	

Initial Requirements	• **Question-Answering (QA) Pipeline:** Implementation of the LogicQA framework to autonomously generate and verify inspection checklists.
	• **Explainability & Localization:** Generation of heatmaps or text-based justifications explaining why a scene is considered anomalous (e.g., "Missing component in compartment B").
	• **Baseline Implementation:** Establish performance benchmarks using both structural (e.g. PatchCore) and semantic (e.g. SALAD) baselines.
	• **Edge AI Deployment:** Test and optimize the VLM-based approach on the NVIDIA Jetson Orin Nano platform.
	
	• **Comprehensive Evaluation:**
	  • Identify and apply suitable metrics: e.g. AUROC, F1-Score
	  • Measure inference latency and throughput for industrial feasibility.
	  • Perform a comparative study between local VLMs (e.g., VILA, Qwen-VL) and cloud-based VLMs (e.g., GPT-4o).

Starting Points	• Dataset: MVTec LOCO AD • LogicQA Paper • Anomalib for structural baselines • SALAD as semantic baseline (example)

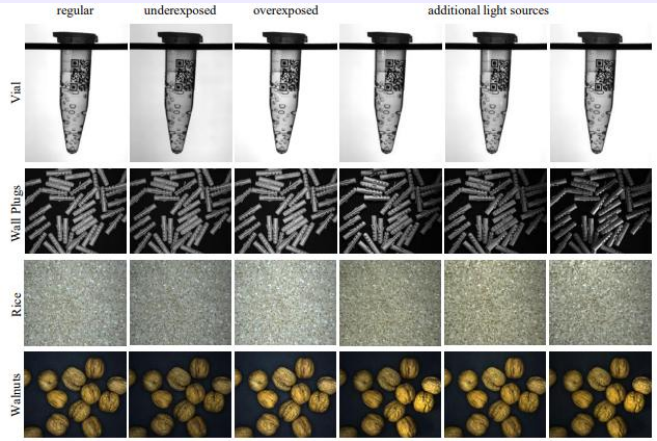


Challenge 4: High Precision AD

Challenge Giver	MVTec (Ulf Schulmeyer)
This challenge addresses the MVTec AD 2 dataset. AD 2 introduces extreme challenges: micro-defects (tiny pixel footprints), transparent/specular objects, and distribution shifts caused by varying illumination (e.g., dark-field vs. bright-field). Teams must develop a robust pipeline capable of maintaining high precision across diverse industrial lighting scenarios.	

Initial Requirements	• **High-Resolution Detail Preservation:** Implement a strategy (e.g., Tiling, Patching, or Multi-Scale approaches) to handle high-resolution images without losing details.
	• **Synthetic Lighting Augmentation:** Develop and integrate augmentation pipelines (utilizing libraries like Kornia/Albumentations or generative style transfer) to make models invariant to lighting shifts.
	• **Hybrid Ensemble Design:** Design an ensemble system that fuses complementary approaches.
	• **Baseline Implementation:** Establish performance benchmarks using structural (e.g. PatchCore) baselines.
	• **Edge AI Deployment:** Test and optimize the VLM-based approach on the NVIDIA Jetson Orin Nano platform.
	• **Comprehensive Evaluation:**
	  • Identify and apply suitable metrics: e.g. AU-PRO@0.05
	  • Measure inference latency and throughput for industrial feasibility.
	  • Perform a comparative study between different ensemble options

Starting Points	• Dataset: MVTec AD 2 • Anomalib for baselines • Image augmentation: Kornia, Albumentations • Inspiration: Challenge winner: Team ISVL



See also: https://sites.google.com/view/vand4-cvpr2026/challenge

Challenge 5: Uncalibrated Monocular VSLAM

Challenge Giver	Markus Friedrich

Professional SLAM systems traditionally require rigid factory calibration. While consumer frameworks like ARCore are stable due to real-time sensor-fusion, they often fail when processing "raw" video files retrospectively. Specifically, they struggle with global metric consistency and lack high-fidelity dense mapping when camera intrinsics are unknown.

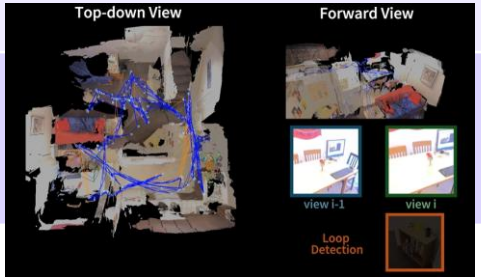
Initial Requirements

Develop a off-device VSLAM pipeline based on existing state-of-the-art monocular dense methods (e.g., ViSTA-SLAM or MASt3R-SLAM). The system must take a smartphone monocular video stream as input, autonomously handle unknown intrinsics, and output a high-precision trajectory and a dense 3D point cloud.

- The system is based on existing monocular VSLAM methods (e.g. ViSTA-SLAM, MASt3R-SLAM)
- **Comprehensive Evaluation:**
    - Identify and apply suitable metrics for the evaluation scenarios (e.g., assessing pose drift and reconstruction fidelity).
    - Evaluate at least two state-of-the-art VSLAM methods
        - Trajectory quality (also vs. ARCore): Benchmarking on the ADVIO dataset and a custom dataset to measure global and local accuracy.
        - 3D point cloud quality (also vs. ARCore): Comparing the native dense output of the VSLAM pipeline against ARCore's mapping results on a self-recorded test dataset.
        - Dataset creation: Develop a custom app for recording high-quality test data (raw video + baseline ARCore logs).
        - Efficiency analysis: Measure the system's latency and memory consumption

Starting Points

- For ground truth 3D point cloud generation: e.g. via colmap + meshroom or colmap + 3DGS
- Point cloud comparison: CloudCompare, metrics e.g. from Open3D
- Trajectory comparison: evo
- Papers: ViSTA-SLAM, MASt3R-SLAM

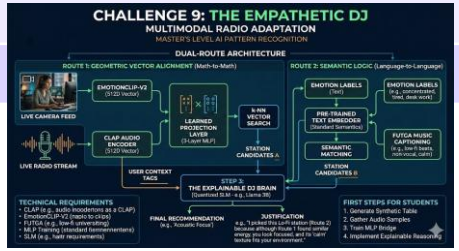


Challenge 6: Multimodal Song Adaptation

Challenge Giver	Markus Friedrich
Build an **Empathetic DJ** that observes a user via camera to extract its emotions and analyzes a live music stream to find the perfect song match.	

Initial Requirements	• Implement a **Dual-Route Architecture:**
	
	1) Align the latent space of the camera with the audio.
	
	  • Use an emotion detector (e.g. EmotionCLIP => emotion embedding), use an audio embedder (e.g. CLAP => audio embedding)
	  • Train a projector network that projects emotion embedding + audio embedding to a single vector via contrastive learning (dataset generation via combination of labeled audio datasets (e.g. PMEmo, DEAM) and labeled video datasets (AffWild2))
	2) Compare descriptions using natural language.
	
	  • Extract textual music description via pre-trained (e.g. FUTGA)
	  • Extract textual emotion description from video (e.g. via EmotionCLIP prompting)
	  • Use pre-trained text embedder to embed the combination of music and emotion description
	• Implement an explainable recommendation system that fetches closest matches from a vector database and returns a textual explanation (refined via local LLM)
	• Implement textual interface also with a feedback channel
	
	Comprehensive Evaluation:
	• Justify model selection for different components based on evaluation for different components (quality, prediction speed, (V)RAM)
	• Compare both routes (1,2) in a user study

Starting Points	• Emotion-CLIP, CLAP, FUTGA

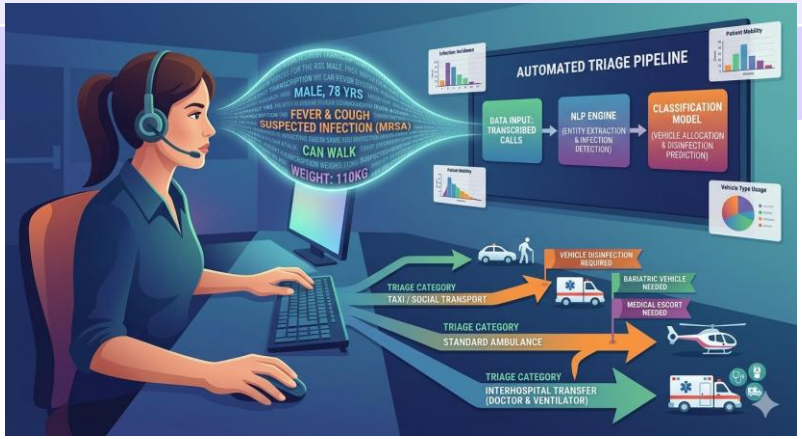


Challenge 7: Automatic Medical Dispatch Triage

Challenge Giver	Feuerwehr München (Lisa Haimerl, Dr. Florian Dax)
Medical dispatchers must distinguish between qualified transports and simple logistics. Misidentifying a patient's needs (e.g., infection status, weight, or mobility) leads to resource waste, such as using an ambulance for a "taxi-level" home transport or failing to account for vehicle disinfection downtime.	

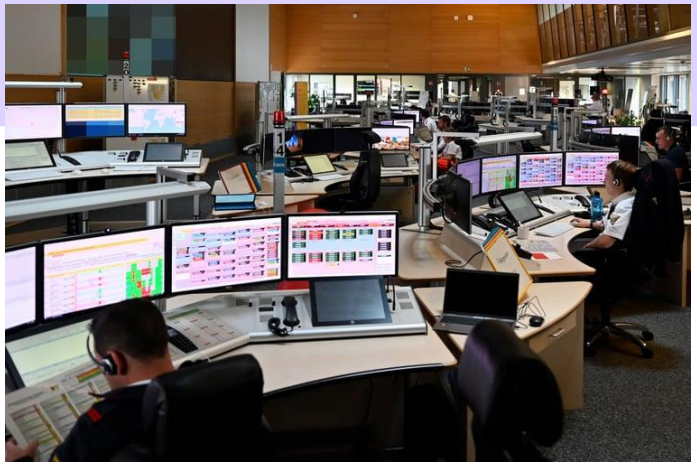
Initial Requirements	Develop an NLP pipeline to analyze transcribed dispatch calls. The system must categorize requests into transport tiers (e.g., Taxi, Social Transport, Ambulance, or Helicopter) and identify critical operational constraints from raw text.
	
	• Automated extraction of 3 core pillars:
	
	  • Logistical: Pickup/Drop-off locations.
	  • Medical: Diagnoses, equipment (ventilators, monitors) and infection status (e.g., MRSA).
	  • Patient-Specific: Mobility status (sitting/lying), cognitive state (dementia/flight risk), and physical dimensions (weight/girth).
	
	• Mapping of extracted features to the correct vehicle/service type.
	
	• Development of a methodology to measure model performance (e.g. with expert-labeling) + comprehensive evaluation

Starting Points	• MedBERT MEDBERT.DE • https://spacy.io/



Challenge 8: Semantic Gap Analysis in Medical Voice Bots

Challenge Giver	Feuerwehr München (Lisa Haimerl, Dr. Florian Dax)
While the current voice bot handles standard transport bookings, a significant gap exists between structured system prompts and the messy reality of human dialogue. To improve the bot's reliability, we must identify recurring "blind spots", critical information items or conversational patterns frequently used by callers but currently missing from the bot's logic and prompting strategy.	
Initial Requirements	Conduct a data-driven gap analysis comparing raw dispatch transcripts to the voice bot's current knowledge base (system prompt). Develop a pipeline to automatically discover recurring medical or logistical items that are currently ignored or misunderstood by the bot. • Apply unsupervised learning (e.g., BERTopic or k-means clustering on embeddings) to identify recurring themes in transcripts that do not map to the existing knowledge base. • Build an interactive dashboard using dimensionality reduction (e.g., t-SNE or UMAP) to visualize "High-Entropy" clusters (conversational areas where transcript and knowledge base differ). • Test and evaluate strategies to close these gaps (e.g., through iterative prompt engineering).
Starting Points	• Specialized encoder: MEDBERT.DE • Topic discovery: BERTopic • Clustering: HDBSCAN • Dim. Reduction: UMAP, t-SNE



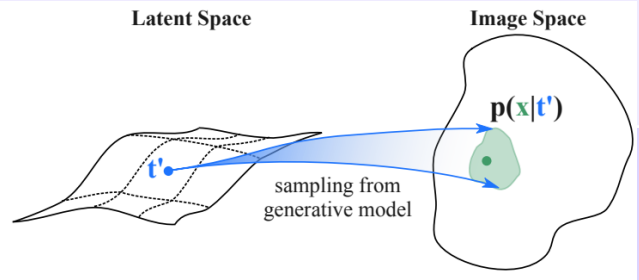
Challenge 9: What's Nice Anyways?

Challenge Giver	Prof. Moritz Schwind (Computational Design For Future Systems, TH Nürnberg)

In professional design and customer pitches, the appeal of an image is often dismissed as "subjective." However, high-end design follows implicit patterns of composition, color, and style. Understanding these patterns algorithmically allows us to move from "random" image generation to a targeted synthesis of "good" or "appealing" design. By analyzing the geometric properties of generative models' latent spaces, we can quantify what makes an image professional and functional for design ideation.

Initial Requirements

- The goal is to analyze a curated dataset of professional design imagery to extract the underlying "feature vectors of appeal." Using the **Hessian Geometry of Latent Spaces**, the project group will explore how specific aesthetic qualities are encoded in models like Stable Diffusion and CLIP, ultimately generating novel images that embody these professional design standards.
- Scraping and preparing a diverse dataset of high-quality design references (e.g., via Behance, Pinterest, or Instagram) to serve as a benchmark for "professional appeal.“
- Representing these images using **CLIP (ViT)** or similar architectures to create a semantic embedding space that links visual features with descriptive design concepts.
- Analyzing the resulting embeddings to identify specific "Design Feature Vectors." Utilizing recent research on the **Hessian Metric** to understand the curvature and structure of the latent space (e.g., identifying phase transitions between "ugly" and "beautiful").
- Developing a method to predict or interpolate novel feature vectors that represent a blend of the input styles.
- Using an image generation pipeline (e.g., Stable Diffusion Image Variation) to synthesize new images based on the predicted vectors and validating their appeal against the original dataset.
- **Expected Outcome:** A diagnostic tool that can quantify the "design quality" of an image and steer generative models toward more professional, pitch-ready results.



Starting Points

- Exemplary Pinterest Collection: https://pin.it/3Y1pjZHyg
- Hessian Geometry of Latent Spaces: https://arxiv.org/pdf/2506.10632
- https://huggingface.co/docs/diffusers/api/pipelines/stable_diffusion/image_variation
- https://openai.com/index/clip/

Fragen?